When is a panel of model judges a valid instrument for human preference?

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PI: [tenure-track faculty, U.S. institution]. Project: https://peereval-ai.github.io. Code: pip install peereval.

The problem

Model judges are replacing human votes faster than anyone has established when they can. Arena's September 2026 study of 1,460 real battles gives the reason for caution: judges picked their own answers 58% of the time against humans' 34%, the three OpenAI judges rated each other 37 points above human ratings, and judges agreed with each other 79.4% of the time but with people only 56.9%. On the 393 battles where all twelve judges agreed, humans concurred 60.6% of the time.

That last number is what matters for panel methods. If model judges share a taste humans do not, averaging more judges buys consensus rather than accuracy, and weighting judges by agreement with the panel entrenches the shared taste instead of correcting it. We built the panel method and tested it against exactly this failure mode. The result contradicted the prior, and the reason we cannot finish the analysis is a data limitation only Arena can remove.

What we have built and measured

PeerEval ranks models by peer review: every model answers the same questions, then judges anonymous pairs of other models' answers in both display orders. First release: 7 models, 3 benchmarks, 137 questions, 40,156 pairwise judgments, as an open-source package with bootstrap intervals and Elo ratings (pip install peereval). We meta-evaluated it against 1,179 real Arena votes (794 crowd, 385 expert), scoring four judges on every battle blind to the human vote. Four findings, all reproducible from the released code.

Self-preference replicates independently. We measure GPT-6 Astra at +54 percentage points of self-preference on our own benchmark set. Arena measures +55.3 on real battles. Two different corpora, two different question distributions, the same number. This is a stable property of the model rather than an artifact of either setup.

Weighting judges by answering skill does not work. PRD Peer Rank assumes a model that answers well also judges well. On our data, contestant skill correlates -0.17 with judging accuracy. The published rule zeroes the weakest contestant, which here is the judge with the best recovery of the human model ranking (Spearman 0.882), while giving a quarter of the vote to the judge that agrees with humans least (0.545 accuracy, +54pp self-preference).

Weighting by inter-judge agreement does work, which contradicts the prior. Replacing the weight signal with agreement among judges, still using no human labels, raises panel agreement with human votes from 0.657 to 0.679 (+0.022, 95% CI [+0.009, +0.034]) and cuts the panel's family bias from +9.2pp to +3.1pp. The gain survives 5-fold cross-fitting and holds when weights are estimated from a corpus disjoint from the battles scored. Given Arena's finding that judges agree with each other far more than with people, agreement weighting should have amplified shared bias. It did the opposite, and we do not yet know why.

The target itself is confounded, which caps every method. Crowd voters prefer the longer answer in 65.8% of battles. A one-line rule, "pick the longer answer," scores 0.658 against the best single judge's 0.690. Searching over all possible weightings of our panel, the ceiling is 0.6958, only +0.006 above using the best judge alone. On this target no aggregation method can demonstrate much, and a judge that resists the verbosity preference is penalised for it: the one judge near length-indifference (44.6%) scores worst on every metric we report.

What we cannot answer without Arena

Three limits are structural rather than effort-limited.

Human position bias is unidentifiable. Each public battle carries one vote in one display order. Our judges are position-debiased by construction, since every pair is judged both ways and averaged, so we score a corrected instrument against a possibly uncorrected label. The marginal split bounds the net effect at roughly plus or minus 3.5pp, but we cannot separate "no bias" from "bias offset by quality imbalance." Per-vote display order would resolve this, and it sets a hard ceiling on how high any judge can score.

Family kinship cannot be controlled. Excluding every model related to one of our judges leaves 92 of 794 battles, and 35 once length is also controlled. A properly powered test needs battles drawn from a pool deliberately disjoint from the judge panel's families.

Panel size. Our strongest results rest on four judges fitting four weights. Agreement-based weighting has more parameters than that can support, and whether the effect holds at eight or twelve judges is the question that decides whether the method is usable.

Proposed work

Aim 1, validity conditions for agreement weighting (months 1-5). Establish when agreement among judges predicts agreement with humans and when it entrenches shared bias. Arena's unanimous-agreement battles are the discriminating case: where all judges concur and humans dissent 39.4% of the time, agreement weighting should fail, and the size of that failure bounds the method's safe operating range. Deliverable: a stated scope condition with a test any practitioner can run on their own panel.

Aim 2, separating human bias from judge bias (months 3-8). With per-vote display order, decompose observed judge-human disagreement into judge position bias (we measure plus or minus 3.2pp), human position bias (currently unmeasurable), verbosity alignment, and residual judgment quality. This yields a corrected ceiling for judge-human agreement, which the field currently treats as 1.0 and which our data suggests is far lower. Deliverable: the ceiling estimate, plus a length- and position-controlled agreement metric.

Aim 3, scaling and replication (months 6-12). Extend from 4 to 12 judges on kin-disjoint battles, pre-registering predictions before running. Deliverable: public replication of Arena's self-preference and family-bias findings on an independent benchmark corpus, and a judged verdict on whether peer weighting is worth its complexity over one-vote-each.

Why this is relevant to Arena

AutoEval issues calibrated ratings from model judgment while human votes accumulate, so its validity rests on exactly the question here: how far a panel of model judges can stand in for the people whose votes it anticipates. Our measured +0.006 weighting ceiling and 65.8% verbosity rate say the honest answer is narrower than current practice assumes, and both are properties of the target rather than any method.

The work also turns Arena's September findings from a caution into a quantity. Self-preference and family bias are documented; what they cost a panel-based rating, and which weighting schemes survive them, is not. We replicate the self-preference result to within 1.3 percentage points on an independent corpus, evidence the effect generalises beyond Arena's distribution. Everything is released, including the negative results: our own published weighting fails, and the best single judge beats our panel on one dataset.

Resources requested, outputs, and budget

Arena datasets. Battle-level records with per-vote display order, model identities, and vote metadata, scoped to a sample sufficient for a kin-disjoint design (target 5,000 battles). Display order is the one input we cannot substitute. Collaboration: review of the validity framework with Arena researchers, and alignment with the self-preference study's methodology so the replication is comparable rather than merely similar. Platform access is optional, for comparing a peer panel against AutoEval on live tasks.

Outputs. Peer-reviewed paper on validity conditions for judge-panel aggregation. Public release of the meta-evaluation harness and all judgments, extending the existing peereval package. A corrected-ceiling estimate for judge-human agreement, and a length and position controlled metric others can adopt. Informal updates to Arena at 6 and 12 months.

Budget, $50,000 over 12 months. Graduate student researcher, partial support, $34,000. Inference and judging API costs, $11,000, sized from measured usage: our four-judge, 1,179-battle meta-evaluation consumed roughly 9,600 API calls, so twelve judges over 5,000 battles in both orders is about 120,000 calls plus the peer-judgment matrix that produces the weights. Conference travel, $5,000. The judging budget is the binding constraint on statistical power.

Stated limitations. Our strongest positive result is significant on one of two datasets, and the agreement-weighting advantage over one-vote-each disappears when battles are restricted to similar-length answers, consistent with part of it being verbosity alignment rather than judgment quality. Three of seven judges were unavailable for the meta-evaluation, so panel results use four. Human position bias remains uncorrected, which is Aim 2.